

Predictive geometry claims are sensitive to the comparator

Post-result sensitivity: all 14 existing features, four recipe baselines and two displacement-conditioned comparators.

	B0 RMSE 2.905 p = 8	B1 RMSE 0.652 p = 11	B2 RMSE 1.391 p = 21	B3 RMSE 1.439 p = 24	B0 + disp. RMSE 1.004 p = 9	B3 + disp. RMSE 1.420 p = 25
Log segment displacement / weight	-0.595*	+0.036	-0.034*	-0.046*	+0.244	-0.045*
Segment stable rank	+0.113	-0.128*	+0.044	+0.158	+0.046	+0.136
Segment entropy rank (top-64)	-0.521*	+0.024	-0.012*	-0.004*	+0.217	-0.002*
Segment row-norm CV	-0.529*	-0.022*	+0.010	+0.009	+0.138	+0.008
Segment top-1% row energy	-0.489*	-0.093*	+0.005	+0.010	+0.028	+0.009
Cumulative displacement / weight	-1.901*	+0.143	+0.017	-0.018*	0.000	0.000
Cumulative stable rank	+0.351	+0.418	+0.052	+0.122	+0.240	+0.142
Segment overlap with AdamW	+0.238	-0.001	+0.148	+0.237	-0.188*	+0.225
Cumulative overlap with AdamW	+0.838	+0.369	+0.233	+0.486	+0.265	-0.085
Exact segment stable rank	+0.105	-0.124*	+0.048	+0.186	+0.053	+0.162
Segment entropy rank (full spectrum)	-0.558*	+0.037	-0.097*	-0.083*	+0.256	-0.072*
Exact cumulative stable rank	+0.354	+0.349	-0.077*	-0.066*	+0.209	-0.040
Exact segment overlap with AdamW	+0.159	-0.050	+0.116	+0.206	-0.164*	+0.175
Exact cumulative overlap with AdamW	+0.406	+0.746	+0.066	+0.701	+0.448	+0.597

1.5
1.0
0.5
0.0
-0.5
-1.0
-1.5
Added feature – comparator RMSE (points)

Negative values favor the added feature. *: lower pooled RMSE and improvements in ≥3/4 held-dose folds.
 B0: original; B1: + raw LR; B2: optimizer × stage + log/raw LR; B3: B2 + nominal cumulative LR budget.
 Same 60 states and training seed. All flags are exploratory, not significance or causal evidence.